

CS/B.Tech/IT/EVEN/SEM-6/IT-603/2015-16



**MAULANA ABUL KALAM AZAD UNIVERSITY OF
TECHNOLOGY, WEST BENGAL**

Paper Code : IT-603

SOFTWARE ENGINEERING

Time Allotted : 3 Hours

Full Marks : 70

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own
words as far as practicable.*

GROUP - A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for the following :

10 × 1 = 10

- i) Which is not a software life cycle model ?
 - a) Spiral model
 - b) Waterfall model
 - c) Prototyping model
 - d) Capability maturity model.
- ii) Project risk factor is considered in
 - a) Spiral model
 - b) Waterfall model
 - c) Prototyping model
 - d) Iterative enhancement model.

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- iii) SDLC stands for
 - a) Software Design Life Cycle
 - b) Software Development Life Cycle
 - c) System Design Life Cycle
 - d) System Development Life Cycle.
- iv) SRS stands for
 - a) Software Requirement Specification
 - b) System Requirement Specification
 - c) Software Requirement Solution
 - d) none of these.
- v) The objective of testing is
 - a) debugging
 - b) to uncover error
 - c) to gain modularity
 - d) to analyze system.
- vi) Which of the following is type of testing ?
 - a) Recovery testing
 - b) Stress testing
 - c) Security testing
 - d) All of these.
- vii) CMM is
 - a) Company Maturity Model
 - b) Capability Maturity Model
 - c) Configuration Maturity Model
 - d) Company Management Model.
- viii) Configuration management is
 - a) Management of the config file of operating system
 - b) Identifying and controlling changes to the configuration of the system
 - c) Process of configuring the management processes and practices
 - d) none of these.

- ix) Which of the following is not a product metric ?
 a) Size b) Reliability
 c) Productivity d) Flexibility.
- x) Effort is measured in terms of
 a) persons b) rupees
 c) month d) persons - month.

GROUP - B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. What is Software Engineering ? What do you mean by feasibility study ?
3. What are the advantages of function points over the size metric of LOC ?
4. Distinguish between static and dynamic testing.
5. What is symbolic execution ? Consider the following function :

```
function max ( x, y, z : integer ) : integer;
begin
  if x ≤ y then
    max = y
  else
    max = x;
  if max < z then
    max = z;
end;
```

Draw a symbolic execution tree for it.

6. Write a comparative study between Waterfall model and Iterative model ?

GROUP - C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. Compare top-down and bottom-up Integration testing. What is acceptance testing ? Define Software 'Reliability' and 'Availability'. What is algorithmic cost estimation ? Consider a project to develop a full screen editor. The sizes for the major modules are estimated to be 4 KLOC, 2 KLOC, 1 KLOC, 2 KLOC and 3 KLOC. Use COCOMO to determine cost and schedule estimates for different phases. Assume that the significant cost drivers adjustment factors to be 1-216.6. $2 + 3 + 3 + 7$
8. What are CMM and COCOMO ? Explain the levels of CMM and COCOMO in brief. Why is Risk Management so important ? $6 + 7 + 2$
9. Explain user interface design ? What is object oriented design ? What do you mean by regression testing ? $7 + 6 + 2$
10. Explain the various steps used to decide the cost of a proposed software system. Explain when we use PART an GANTT ? Suppose we are developing software and expect to have about 3,00,000 lines of code. Compute the effort and the development time for each of the organic and embedded development mode. $5 + 10$
11. Explain program evaluation and review technique in detail. Describe SRS. Differentiate between use case and class diagram in SPM. $10 + 5$